ENGINEERING CHANGE NOTE (ECN)

PROJECT: PROJECT-CE
ORIGINATION: Prof. Kidd
PURPOSE OF CHANGE: Addition of a PWM-driven motor fan and push-button state machine.
DOCUMENT NUMBER: ECN 0.3.1 (revised 4/1/2030 - kidd)
DATE OF PUBLICATION: October 24, 2026

LEAD ENGINEER: Kid Kidd, UMBC
TEAM LEAD: Great Akinrotimi

EXECUTIVE SUMMARY: This engineering change modifies the PROJECT-MYMILLI design to incorporate a DC motor-driving fan with four operating states: OFF, LOW, MEDIUM, and HIGH. The speed of the fan is controlled by a PWM signal and determined by a momentary push button that cycles through the states sequentially. The existing Cyclic Executive task manager and custom myMilli() timebase will be preserved, ensuring the new button and motor tasks operate asynchronously without affecting the previously implemented LED and serial I/O functionalities.

HIGH-LEVEL DESCRIPTION: Modify the design of PROJECT-MYMILLI to add a motor driving a fan with 4 possible states, cycled via a push button.

REASON FOR CHANGE: The customer has added the new requirement of having a cooling fan added to the device.

Changed Documents: 10/24/26 (SDD.4.1) System Design Document, updated to include Customer Requirements C6-C12 and High-Level Requirements HL.7-HL.18.

10/24/26 (TVR.4.1) Testing and Validation Requirements, updated to include Serial I/O dialog (Table 4) and Button/Fan state sequences (Table 5), alongside justified CFM/dB calculations for EC.1.3.

Added Documents: None

Removed Documents: None

Replaced Documents: None

Any Additional Design Considerations:
CUSTOMER REQUIREMENTS: The customer requirements, C1 through C5, are identical to that of PROJECT-ASYNC. See [1] CUSTOMER REQUIREMENTS section.

HIGH-LEVEL TECHNICAL REQUIREMENTS: The high-level technical requirements, HL.1 through HL.3, are identical to that of PROJECT-ASYNC. See [1] HIGH-LEVEL TECHNICAL REQUIREMENTS section.

REFERENCES: [1] Reference document Great Akinrotimi Project#1 Asynchronous Lab Report.pdf

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NOTE: The design documents that have been modified must have any deleted text struck out along with the date, and any addi0ons made along with the date. If an en0re document is removed or replaced, a note to that effect has to be added to the ECN. If document(s) are added, a note to that effect has to be added to the ECN.